



Short Communication

Long-term effects of grass roots on gas permeability in unsaturated simulated landfill covers

Jun Jun Ni *, Charles Wang Wai Ng

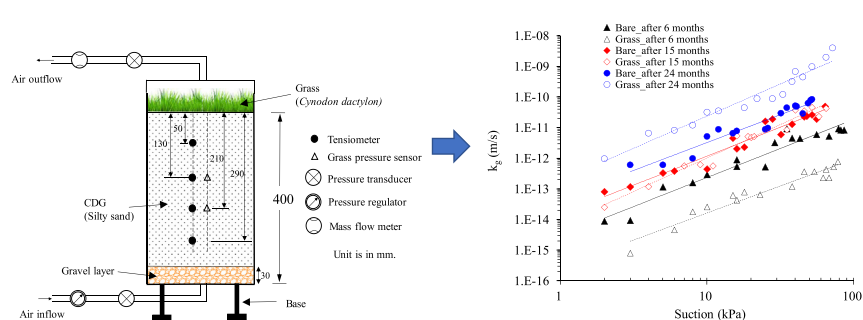
Department of Civil and Environmental Engineering, the Hong Kong University of Science and Technology, Hong Kong Special Administrative Region



HIGHLIGHTS

- Gas permeability increased linearly with suction in semi-log scale.
- Plant age has significant effect on gas permeability.
- After 6 months, gas permeability in grass-covered soil was smaller than bare soil.
- After 2 years, gas permeability of grass-covered soil was higher than bare soil.

GRAPHICAL ABSTRACT



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ABSTRACT

Landfill cover is a barrier that can reduce landfill gas emission. Vegetation could affect soil hydraulic properties of landfill cover, while its effects on gas permeability is not well understood. This paper investigated the effects of plant roots on the evolution of gas permeability (k_g) in unsaturated landfill cover over a two-year period. One grass species (*Cynodon dactylon*) was selected for testing in the laboratory and there were six replications. Bare soil was used as a reference. k_g was determined after 6, 15 and 24 months of grass growth. The test results show that with the increase of suction, k_g increased linearly in log scale for both bare and grass-covered soils. k_g of grass-covered can be lower or higher than that of bare soil, depending on the plant age. After 6 months, k_g of grass-covered soil was smaller than that of bare soil by 85%, due to the root occupancy of soil pore space. However, after 15 and 24 months, k_g of grass-covered soil increased by 2–3 orders of magnitude within the root zone over the suction range considered (2–86 kPa). After 2 years, k_g of grass-covered soil was up to one order of magnitude higher than that in bare soil. It implies that mature vegetation is not beneficial for reducing gas emission in final landfill covers.

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1. Introduction

Landfill gas that contains CH_4 , CO_2 has a number of negative environmental impacts, such as increasing the climate temperature and surrounding air pollution (Wickramarachchi et al., 2011; Feng et al.,

2018). Final covers, which act as barriers between municipal solid waste and the atmosphere, have been widely used to reduce water infiltration and gas emission (Scanlon et al., 2005; Ng et al., 2016). The low gas permeability reduces the availability of CH_4 and O_2 and hence methane oxidation. Laboratory tests showed that gas permeability decreased with increasing water content and compacting dry density, while increased with larger pore size (Wickramarachchi et al., 2011; Hamamoto et al., 2011). When soil moisture content decreased below

* Corresponding author.

E-mail address: jn1aa@connect.ust.hk (J.J. Ni).

5%, the air-filled void may not be interconnected and gases have to diffuse in soil water (Scheutz and Kjeldsen, 2004), resulting in dramatical reduction in biological activity and methane oxidation.

Vegetation is commonly transplanted upon final covers. Recent field and laboratory studies found that in addition to its aesthetic contribution, vegetation also changes soil hydraulic properties via root occupancy of soil pore space (Ng et al., 2016; Ni et al., 2017). Changes in soil pore structure cause variations of soil pore size distributions and hence soil water retention ability and water permeability (Scanlan, 2009; Leung et al., 2015). Plant roots do not always increase or decrease water permeability/infiltration, depending on plant type (Rahardjo et al., 2014), plant spacing (Ng et al., 2016) and plant age (Jotisankasa and Sirirattanachai, 2017). It should be noted that decay roots induced formation of soil macro-pores (i.e., increase in void ratio) could reduce soil water retention and increase water permeability (Ghestem et al., 2011). It can be deduced that root-induced those changes in soil pore structures may also potentially affect gas permeability, especially when the soil is under unsaturated state. Indeed, Zhan et al. (2016)

found that in loosely compacted silt in field, the growth of plant roots in the short term tended to fill the large pores, resulting in a decrease in gas permeability by one order of magnitude. However, variations of gas permeability considering growth of plant roots in the long term are still not clear yet.

This study aims to investigate the evolution of gas permeability in unsaturated vegetated soils considering the growth plant roots. A commonly found grass species (*Cynodon dactylon*) was selected for investigation. There were six replications for the grass case. One bare case was tested for the reference. Gas permeability measurement was conducted at different growing stages (i.e., 6, 15 and 24 months).

2. Materials and methods

2.1. Test setup and instrumentation

There were 6 columns for grass-covered soil (Fig. 1(a)) and one for bare soil. Each column had a height of 400 mm and a diameter of

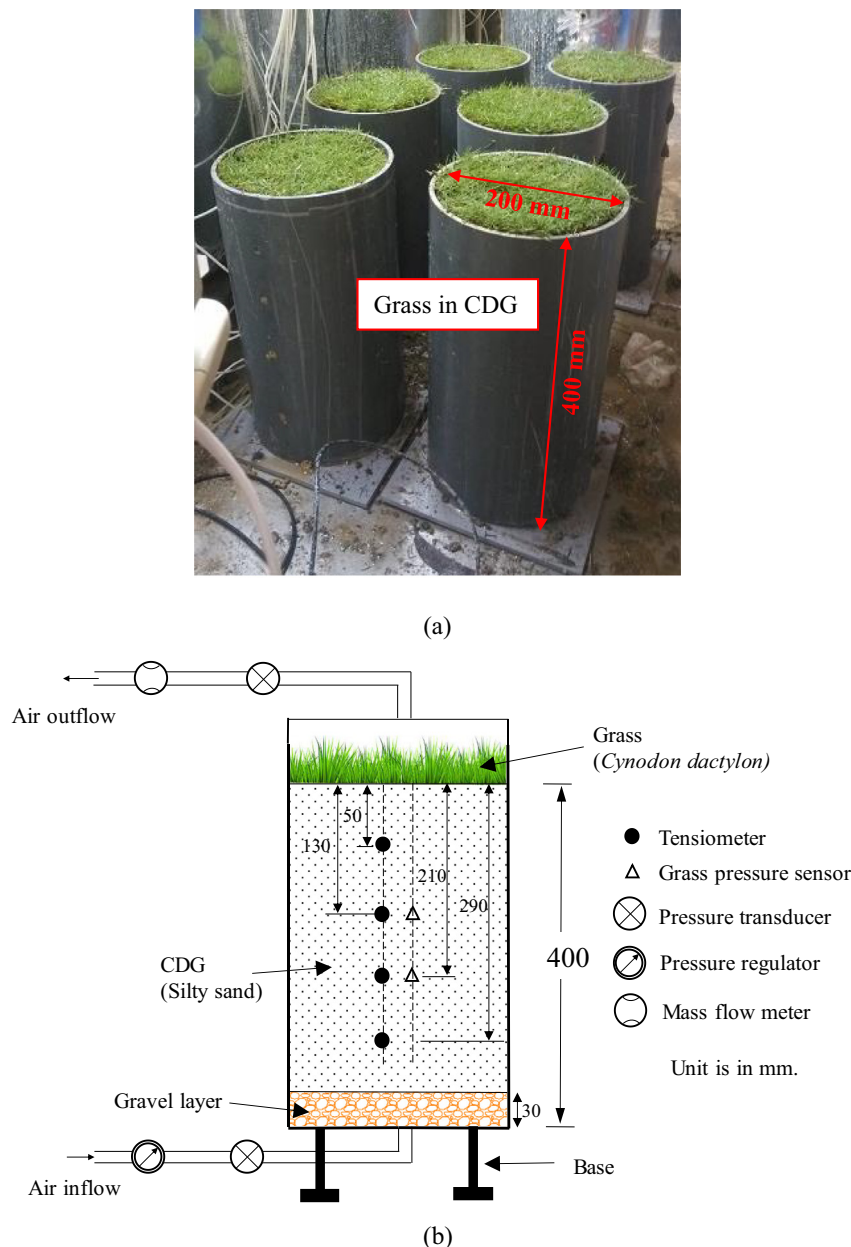


Fig. 1. (a) Overview of the columns vegetated with grass; (b) test setup for the gas permeability test.

200 mm. There were drainage holes with a diameter of 5 mm at the bottom of each column to allow for free water drainage and gas influx during testing. All columns were placed in a plant room with relative humidity $60 \pm 5\%$ and air temperature $25 \pm 1^\circ\text{C}$. Cool white fluorescent lamps were used to provide a constant radiant energy of $5.0 \text{ (MJ/m}^2\text{) / day}$ for facilitating plant growth (Ng et al., 2016).

An array of miniature-tip tensiometers (2100 F, Soil Moisture Equipment Cooperation) were installed at 50, 130, 210 and 290 mm depth of each column to measure negative pore water pressure or matric suction up to 90 kPa (Fig. 1(b)). At the depth of 130 and 210 mm, two gas pressure transducers (Chen, 2016) were installed to measure the gas pressure during testing. Depending on the gas flow rate, two types of air flow meter with different ranges ($0\text{--}100 \text{ mL/min}$ and $0\text{--}30 \text{ L/min}$) were used. The accuracies were 0.5 mL/min and 5 mL/min , respectively.

2.2. Soil type and sample preparation method

The soil compacted in each column was completely decomposed granite (CDG), which is commonly found in tropical and subtropical areas. The contents of gravel, sand, silt and clay of CDG are 19, 42, 27 and 12%, respectively. The plastic limit is 26% and the liquid limit is 44%. Hence, the plasticity index was 18%. According to the Unified Soil Classification System (USCS; ASTM, 2010), the CDG is classified as silty sand (SM). The maximum dry density and corresponding optimum gravimetric water content are 1870 kg/m^3 and 12% respectively. Nutrient tests following the methods suggested by Sparks et al. (1996) and Buhrke et al. (1998) show that the contents of N and P in the CDG are 0.76 and 0.53 mg/kg, respectively. The contents of total C, K and Ca are 1.2%, 7.0% and 0.96%, respectively. Other index properties are summarized in Table 1.

Relative compaction (RC) of 95% was selected in this study. Such a RC is normally adopted for landfill cover design in Hong Kong (GEO, 2011). Both laboratory and field tests have shown that the species selected in this study, *Cynodon dactylon*, is able to survive under this level of compaction (Ng et al., 2014; Ni et al., 2017). In each column, soil was compacted in 12 layers, with each layer a height of 30 mm. Soil surface was scarified to provide better contact between each successive layers.

2.3. Selected plant species

A grass species (*Cynodon dactylon*, also known as Bermuda grass) were selected for testing. It was selected because it is ecologically

suitable for rehabilitation and restoration of landfill cover in many parts of Asia (GEO, 2011). Before transplantation, grass turf with shoot length of $50 \pm 6 \text{ mm}$ (mean $\pm$ standard error of mean) and root depth of $40 \pm 8 \text{ mm}$ was provided by Tung Kee Garden Horticulture Ltd. in Hong Kong. When transplanting grass turf, surface soil was scarified to ensure the better contact between grass roots and soil. After transplantations, the plants were left grown in the plant room for two years. During this period, all grass-covered and bare columns were irrigated every 3 days unless stated otherwise. In order to observe the growth of grass roots, soil cores (76 mm in diameter and 100 mm in height) were sampled with a cylindrical core cutter after the test.

2.4. Testing procedures

During the two-year period, gas permeability was measured after grass growth for 6, 15 and 24 months respectively. Before gas permeability measurement, surface of all planted and bare columns was ponded with water until basal percolation was observed and suction at all instrumented depths became zero. Then, all columns were left in the plant room for evapotranspiration. Gas permeability was measured continuously during the drying period. The gas selected was compressed CO_2 . The bottom of each column was connected to a gas chamber, which can provide constant gas pressure of 20 kPa (a typical landfill gas pressure; Chen, 2016). A gas flow meter was mounted on the top to measure the gas flow rate. Darcy's law is used to calculate the advective flow velocity for any fluid (gas or liquid) in response to a pressure gradient (Hubbert, 1957). According to Darcy's law, flow velocity is expressed as

$$v = \frac{q}{A} = k_g i \quad (1)$$

where k_g is gas permeability, i is gas pressure gradient, q is flow rate, A is soil cross sectional area. $i = \frac{\Delta h}{L}$, where Δh is gas pressure difference and L is length of soil. Therefore

$$\Delta q = k_g \frac{\Delta p}{\gamma L} A \text{ or } k_g = \frac{\Delta q \gamma}{\Delta p A} L \quad (2)$$

where Δp is pressure difference and γ is specific weight of flow medium.

Table 1
Index properties of completely decomposed granite (CDG).

Index properties	Value
Standard compaction tests	
Maximum dry density: kg/m^3	1870
Optimum moisture content: %	12
Particle-size distribution	
Gravel content ($>2 \text{ mm}$): %	19
Sand content ($\leq 2 \text{ mm}$): %	42
Silt content ($\leq 63 \mu\text{m}$): %	27
Clay content ($\leq 2 \mu\text{m}$): %	12
D_{10} : mm	0.15
D_{30} : mm	0.7
D_{60} : mm	2
Coefficient of uniformity (D_{60}/D_{10})	13.3
Coefficient of curvature ($(D_{30})^2/(D_{60}D_{10})$)	1.6
Specific gravity	2.60
Atterberg limit	
Plastic limit: %	26
Liquid limit: %	44
Plasticity index: %	18
Unified Soil Classification System (USCS) ^a	Silty sand (SM)

^a ASTM (2010).

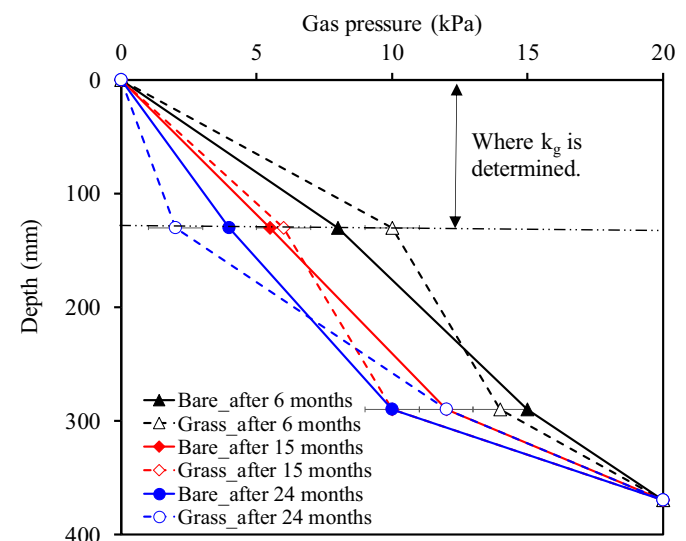


Fig. 2. Gas pressure distributions at similar suction values (52–56 kPa at depth of 50 mm).

3. Interpretation of test results

3.1. Gas pressure distribution

Fig. 2 shows the gas pressure distribution at similar suction values (52–56 kPa at the shallow depth) for bare and grass-covered soils. For the top and bottom boundaries, gas pressures were 0 and 20 kPa, respectively. Gas pressure decreased gradually along depth inside columns. After 6 months of grass growth, gas pressure difference at the shallow depth (i.e., 130 mm) of grass-covered soil was larger than that in the bare soil. High pressure difference corresponds to the high pressure gradient, indicating high resistance to gas flow at a given gas flow rate. It has been found that plant roots tend to grow in large pores, where they access water, air and mineral nutrients (Stewart et al., 1999). Root occupancy of large pores is a key factor affecting the gas permeability of soil (Moldrup et al., 2001). Besides, during root penetration, soil surrounding plant roots can be densified, which also reduced large pore size.

Interestingly, with the grass growth, gas pressure gradient at the shallow depth decreased for both bare and grass-covered soils. However, the decreased amount was much more significant for the grass-covered case. Reduction of gas pressure gradient in the bare soil was probably due to the small cracks observed during wetting and drying cycles. Those crack channels would facilitate gas flow even under small pressure gradient. In grass-covered soil, the presence of decay roots after 15 months' growth created preferential channels within the root zone. Those channels became the main gas flow paths during gas migration. This phenomenon was much more obvious after 24 months' grass growth. Decay roots induced preferential channels were also observed in mixed species and closely-spacing planting (Ng et al., 2016; Ni et al., 2017).

3.2. Variations of gas permeability with time

Fig. 3 shows the gas permeability (k_g) with time for both bare and grass-covered soils. The embedded photo shows the growth condition of grass roots at shallow depth of 100 mm after the test. It consistently shows that k_g increased with time for both cases. k_g of bare soil increased by almost 15 times in the shallow depth, while k_g of grass-covered soil increased by around 400 times. It is found that k_g of the vegetated soil can be higher or lower than that of bare soil, depending on the plant age. After 6 months, k_g of grass-covered soil was smaller than that of bare soil by 85%. On the contrary, k_g of grass-covered soil

was larger than that of bare one by 140% and 310% after 15 and 24 months of growth, respectively.

On one hand, it is not uncommon to find suction induced crack propagation in compacted soils (Bordoloi et al., 2018). With the presence of grass roots, higher suction was induced during drying even below the root zone, which would facilitate crack development. It has been reported that roots can proliferate in cracks (Ghestem et al., 2011) and further exacerbate plane of weakness where cracks propagate. Those enhanced cracks by plant roots would increase gas permeability significantly. On the other hand, as mentioned in previous section, decay roots caused by plant aging can also contribute to the increase k_g in Fig. 3. Based on the field and laboratory data, it has been found a root-induced increase in saturated water permeability by 1.3–6.5 times due to root decaying/aging (Li and Ghodrati, 1994; Rasse et al., 2000; Scanlan, 2009; Vergani and Graf, 2015). The increased water permeability was closely related to the decay roots-induced macro-pores and preferential channels, which also increased gas flow when the soils are partially filled with water.

3.3. Gas permeability function

Fig. 4 shows the relationship between gas permeability (k_g) and soil suction for both bare and grass-covered conditions after 6, 15 and 24 months, respectively. This relationship is also called gas permeability function. Test data in each scenario was best fitted with the least square method. It can be found that for the suction range considered (2–86 kPa), there existed linear relationships between k_g and suction for all the scenarios. k_g increased with suction increase. Besides, the slope of fitted lines were almost the same. After 6 months of growth, k_g in grass-covered soil was about one order of magnitude smaller than that in bare soil. Similar observations were also captured in compacted silty clay with mixture of *Ilex purpurea* Hassk, ryegrass and bluegrass (Zhan et al., 2016).

After continuous growth of grass, k_g -suction relationship for both bare and vegetated soils shifted upwards in Fig. 4. That means k_g increased at any specific suction after 6 months. However, the increased amount in vegetated case was much larger than that in bare soil by up to two orders of magnitude. On one hand, the higher k_g in vegetated soils can increase the available CH_4 and O_2 in the soil and hence enhance the methane oxidation in landfill cover. On the other hand, it may also increase gas emission into atmosphere. In order to balance the ecological restoration using vegetation and the practical function of landfill cover in reducing landfill gas emission, materials with low

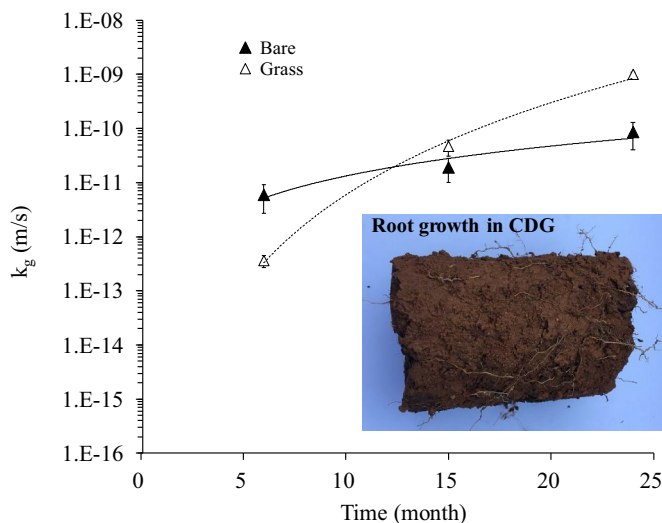


Fig. 3. Variations of gas permeability with time at similar suction values (52–56 kPa at depth of 50 mm).

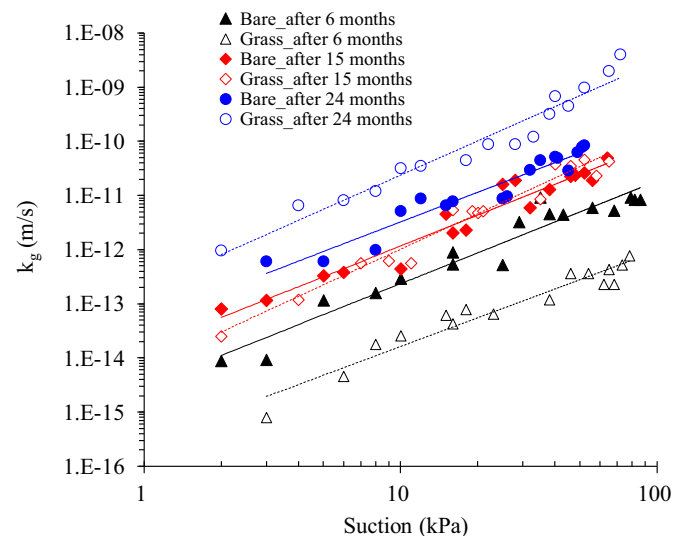


Fig. 4. Comparison of gas permeability function in bare and grass-covered soils.

permeability (such as membranes, compacted clay) should be included at the bottom of traditional landfill covers.

4. Conclusions

This study investigated the influence of grass roots on the evolution of gas permeability (k_g) in unsaturated landfill covers over two-year period. k_g was measured in soil columns by applying constant gas pressure at the bottom. The results were interpreted with Darcy's law. It was found that with the increase of suction, k_g increased linearly in log scale for both bare and grass-covered soils. At the early stage (i.e., 6 months), k_g in grass-covered soil was smaller than that in bare soil by 85%. This is due to the blockage of flow channels by plant roots. After that, k_g of bare and vegetated soils increased, but the increased amount in grass-covered soil after 24 months was two orders of magnitude higher than that of bare soil over a wide suction range (2–86 kPa). Decay roots associated with plant aging can cause preferential channels and hence increase the gas migration in the soil. It implies that even though vegetation is beneficial to ecological restoration and aesthetical views, it also has negative effect on the reduction of gas emission in landfill final covers.

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References

- ASTM, 2010. Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System). American Society for Testing and Materials, West Conshohocken, PA, USA.
- Bordoloi, S., Garg, A., Sreedeeep, S., Peng, L., Mei, G., 2018. Investigation of cracking and water availability of soil-biochar composite synthesized from invasive weed water hyacinth. *Bioresour. Technol.* 263, 665–677.
- Buhrke, V.E., Jenkins, R., Smith, D.K., 1998. A Practical Guide for the Preparation of Specimens for XRF and XRD Analysis. Wiley 0-471-19458-1 ISBN.
- Chen, Z.K., 2016. Gas Breakthrough and Emission in Unsaturated Landfill Final Cover Considering Cracking Effect. PhD Thesis. The Hong Kong University of Science and Technology, Hong Kong.
- Feng, S., Liu, H.W., Chiu, C.F., Ng, C.W.W., 2018. A steady-state analytical profile method for determining methane oxidation in landfill cover. *Sci. Total Environ.* 646, 1528–1535.
- GEO (Geotechnical Engineering Office), 2011. Technical Guidelines on Landscape Treatment for Slopes. Geotechnical Engineering Office, Hong Kong, China.
- Ghestem, M., Sidle, R.C., Stokes, A., 2011. The influence of plant root systems on subsurface flow: implications for slope stability. *Bioscience* 61 (11), 869–879.
- Hamamoto, S., Moldrup, P., Kawamoto, K., Wickramarachchi, P.N., Nagamori, M., Komatsu, T., 2011. Extreme compaction effects on gas transport parameters and estimated climate gas exchange for a landfill final cover soil. *J. Geotech. Geoenviron.* 137 (7), 653–662.
- Hubbert, M.K., 1957. Darcy's law and the field equations of the flow of underground fluids. *Hydrol. Sci. J.* 2 (1), 23–59.
- Jotisankasa, A., Sirirattanachai, T., 2017. Effects of grass roots on soil-water retention curve and permeability function. *Can. Geotech. J.* 54 (11), 1612–1622.
- Leung, A.K., Garg, A., Co, J.L., Ng, C.W.W., Hau, B.C.H., 2015. Effects of the roots of *Cynodon dactylon* and *Shefflera heptaphylla* on water infiltration rate and soil hydraulic conductivity. *Hydrol. Process.* 29 (15), 3342–3354.
- Li, Y., Ghodrati, M., 1994. Preferential transport of nitrate through soil columns containing root channels. *Soil Sci. Soc. Am. J.* 58, 653–659.
- Moldrup, P., Olesen, T., Komatsu, T., Schjønning, P., Rolston, D.E., 2001. Tortuosity, diffusivity, and permeability in the soil liquid and gaseous phases. *Soil Sci. Soc. Am. J.* 65 (3), 613–623.
- Ng, C.W.W., Leung, A.K., Woon, K.X., 2014. Effects of soil density on grass induced suction distributions in compacted soil subjected to rainfall. *Can. Geotech. J.* 51 (3), 311–321.
- Ng, C.W.W., Ni, J.J., Leung, A.K., Zhou, C., Wang, Z.J., 2016. Effects of planting density on tree growth and induced soil suction. *Géotechnique* 66 (9), 711–724.
- Ni, J.J., Leung, A.K., Ng, C.W.W., 2017. Investigation of plant growth and transpiration-induced suction under mixed grass tree conditions. *Can. Geotech. J.* 54 (4), 561–573.
- Rahardjo, H., Satyanaga, A., Leong, E.C., Santoso, V.A., Ng, Y.S., 2014. Performance of an instrumented slope covered with shrubs and deep-rooted grass. *Soils Found.* 54 (3), 417–425.
- Rasse, D.P., Smucker, A.J.M., Santos, D., 2000. Alfalfa root and shoot mulching effects on soil hydraulic properties and aggregation. *Soil Sci. Soc. Am. J.* 64 (2), 725–731.
- Scanlan, C.A., 2009. Processes and Effects of Root-Induced Changes to Soil Hydraulic Properties. PhD Thesis. University of Western Australia, Australia.
- Scanlon, B.R., Reedy, R.C., Keese, K.E., Dwyer, S.F., 2005. Evaluation of evapotranspirative covers for waste containment in arid and semiarid regions in the southwestern USA. *Vadose Zone J.* 5 (2), 55–71.
- Scheutz, C., Kjeldsen, P., 2004. Environmental factors influencing attenuation of methane and hydrochlorofluorocarbons in landfill cover soils. *J. Environ. Qual.* 33, 72–79.
- Sparks, D.L., Page, A.L., Dalton, P.A., Loeppert, R.H., Soltanpour, P.N., Tabata, M.A., Johnston, C.T., 1996. Methods of soil analysis: part 3. Chemical Methods. Soil Science Society of America, American Society of Agronomy, Madison, Wisc.
- Stewart, J.B., Moran, C.J., Wood, J.T., 1999. Macropore sheath: quantification of plant root and soil macropore association. *Plant Soil* 211 (1), 59–67.
- Vergani, C., Graf, F., 2015. Soil permeability, aggregate stability and root growth: a pot experiment from a soil bioengineering perspective. *Ecohydrology* <https://doi.org/10.1002/eco.1686>.
- Wickramarachchi, P., Kawamoto, K., Hamamoto, S., Nagamori, M., Moldrup, P., Komatsu, T., 2011. Effects of dry bulk density and particle size fraction on gas transport parameters in variably saturated landfill cover soil. *Waste Manag.* 31 (12), 2464–2472.
- Zhan, L.T., Qiu, Q.W., Xu, W.J., Chen, Y.M., 2016. Field measurement of gas permeability of compacted loess used as an earthen final cover for a municipal solid waste landfill. *J. Zhejiang Univ. Sci. A* 17 (7), 541–552.